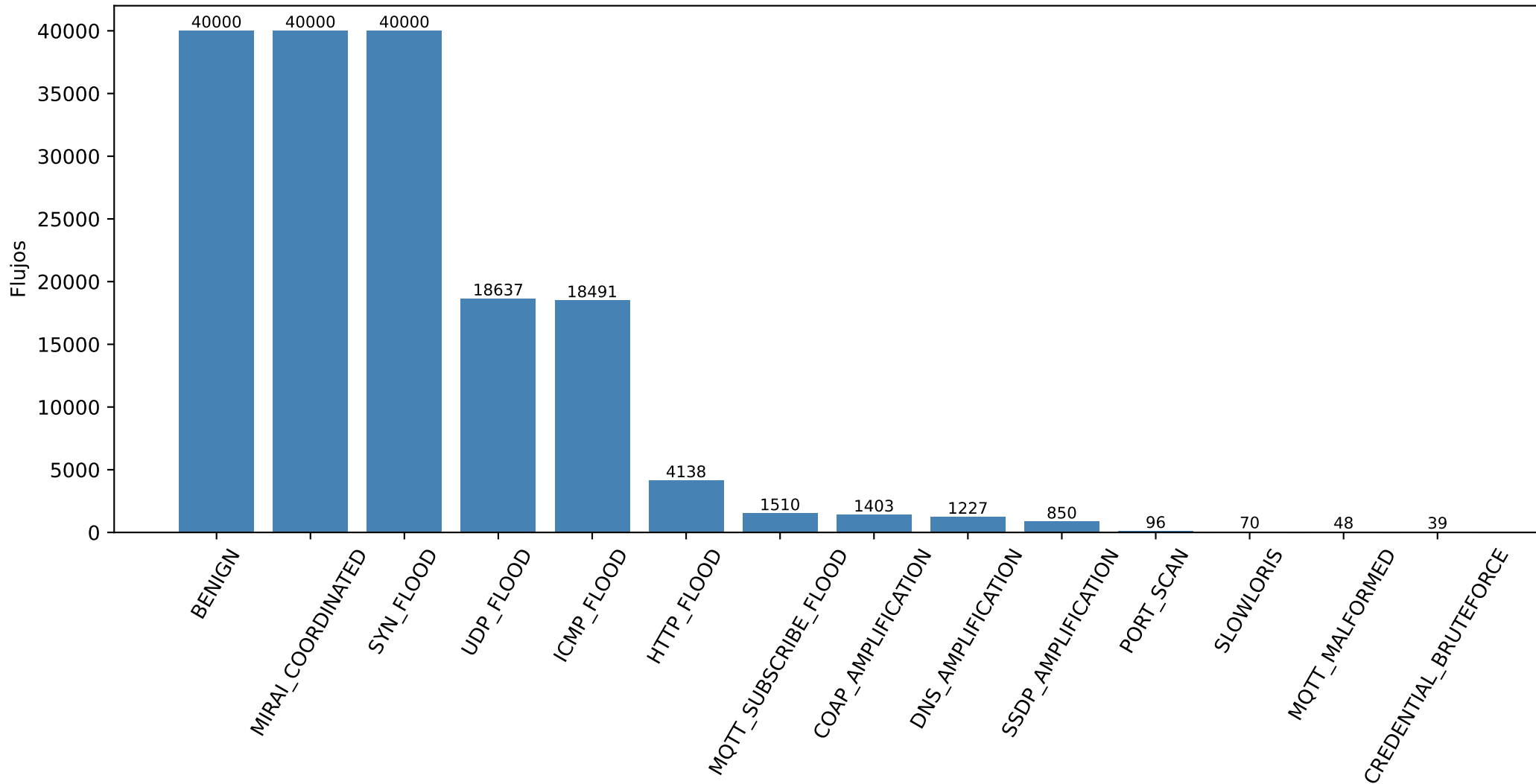


IoT-SDN-AI — Reporte de evaluación

run: run_20260921-135634

Dataset:	208,137 flujos
Features:	46
Clases:	14 (BENIGN, COAP_AMPLIFICATION, CREDENTIAL_BRUTEFORCE, DNS_AMPLIFICATION, HTTP_FLOOD...)
Modelos:	RandomForest, XGBoost, MLP, IsolationForest, AutoEncoder
Estrategia:	stratified train/test split 80/20, scaler StandardScaler

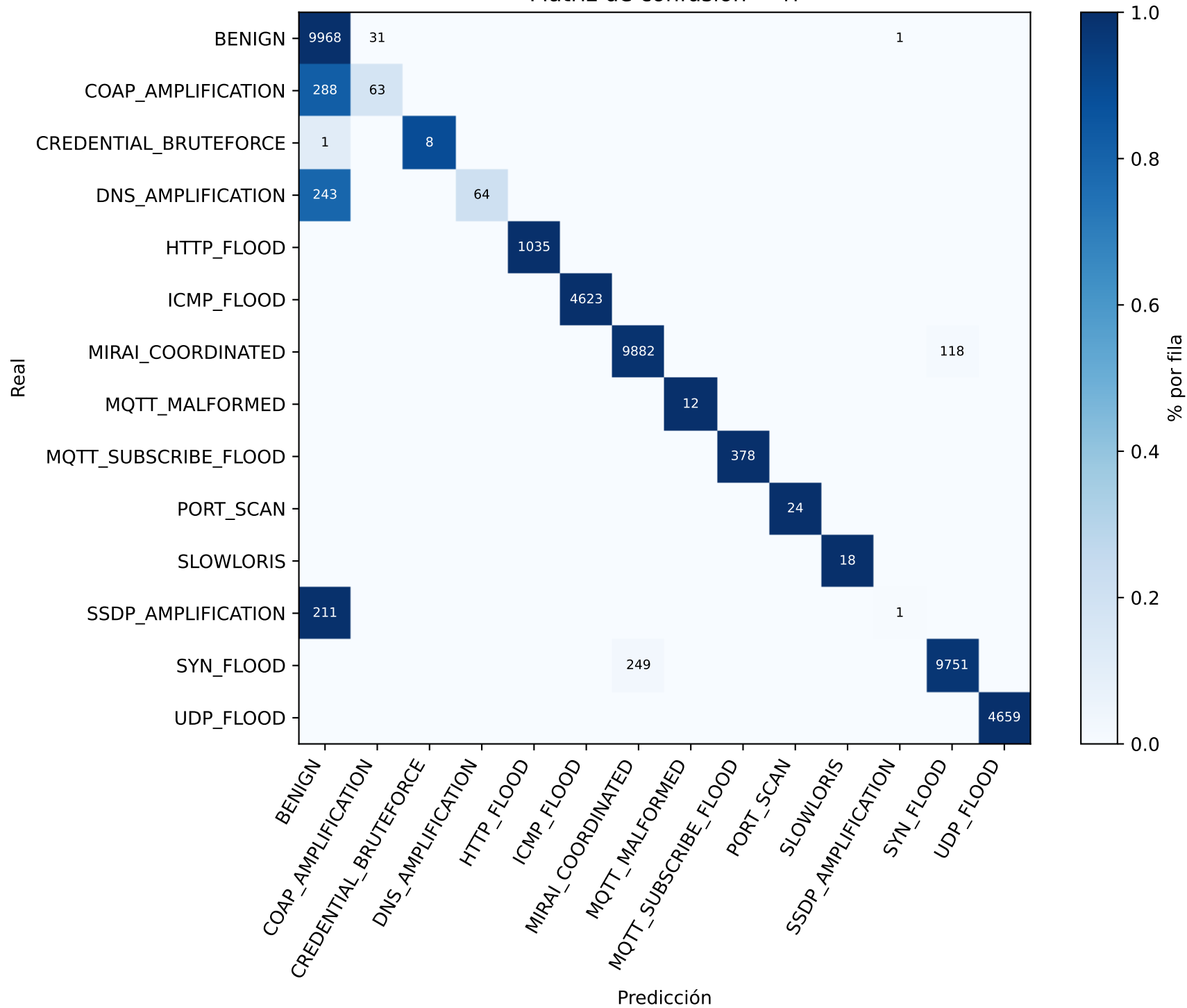
Distribución de clases en el dataset



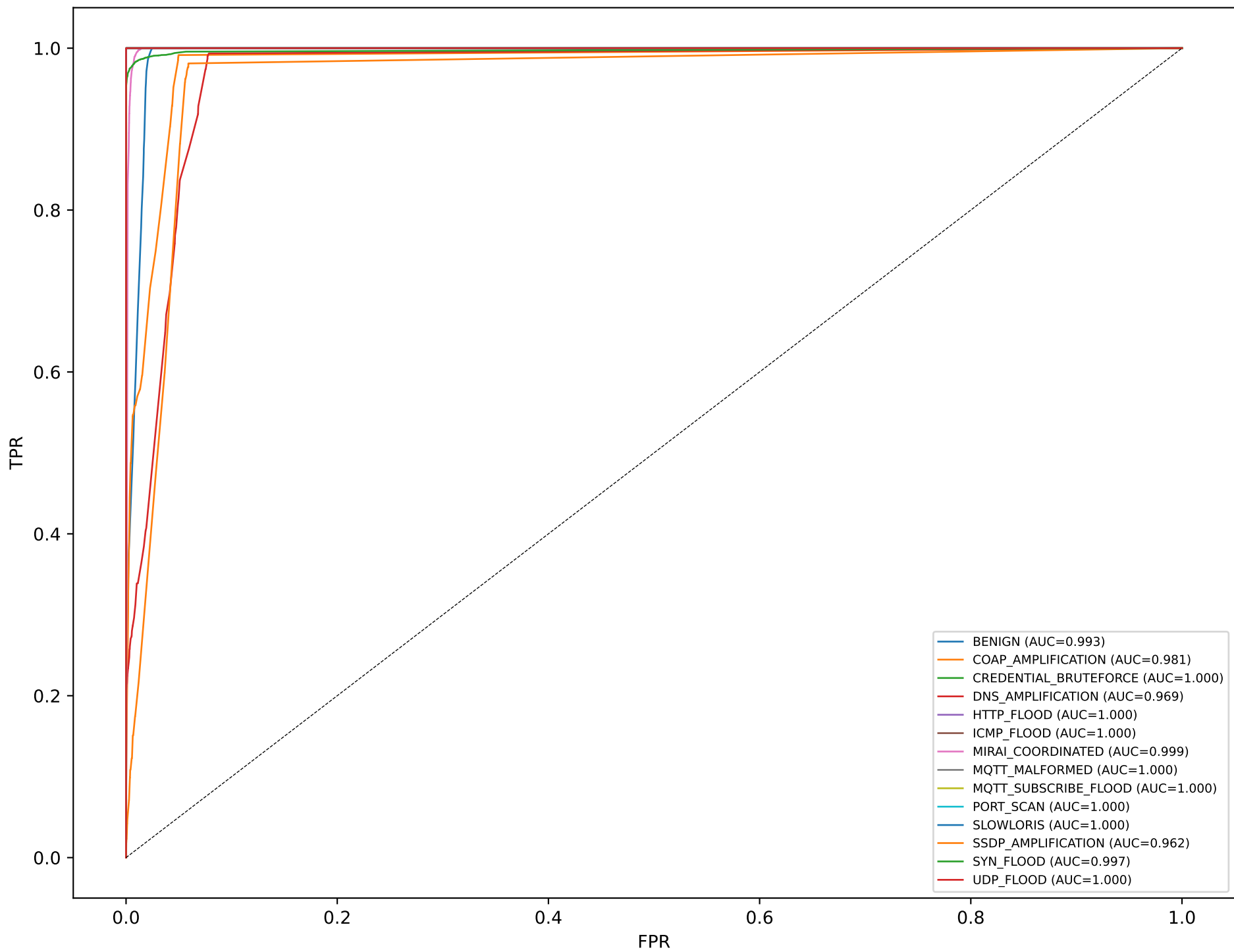
Classification report — rf

	precision	recall	f1-score	support
BENIGN	0.93	1.00	0.96	10000
COAP_AMPLIFICATION	0.67	0.18	0.28	351
CREDENTIAL_BRUTEFORCE	1.00	0.89	0.94	9
DNS_AMPLIFICATION	1.00	0.21	0.35	307
HTTP_FLOOD	1.00	1.00	1.00	1035
ICMP_FLOOD	1.00	1.00	1.00	4623
MIRAI_COORDINATED	0.98	0.99	0.98	10000
MQTT_MALFORMED	1.00	1.00	1.00	12
MQTT_SUBSCRIBE_FLOOD	1.00	1.00	1.00	378
PORT_SCAN	1.00	1.00	1.00	24
SLOWLORIS	1.00	1.00	1.00	18
SSDP_AMPLIFICATION	0.50	0.00	0.01	212
SYN_FLOOD	0.99	0.98	0.98	10000
UDP_FLOOD	1.00	1.00	1.00	4659
accuracy			0.97	41628
macro avg	0.93	0.80	0.82	41628
weighted avg	0.97	0.97	0.97	41628

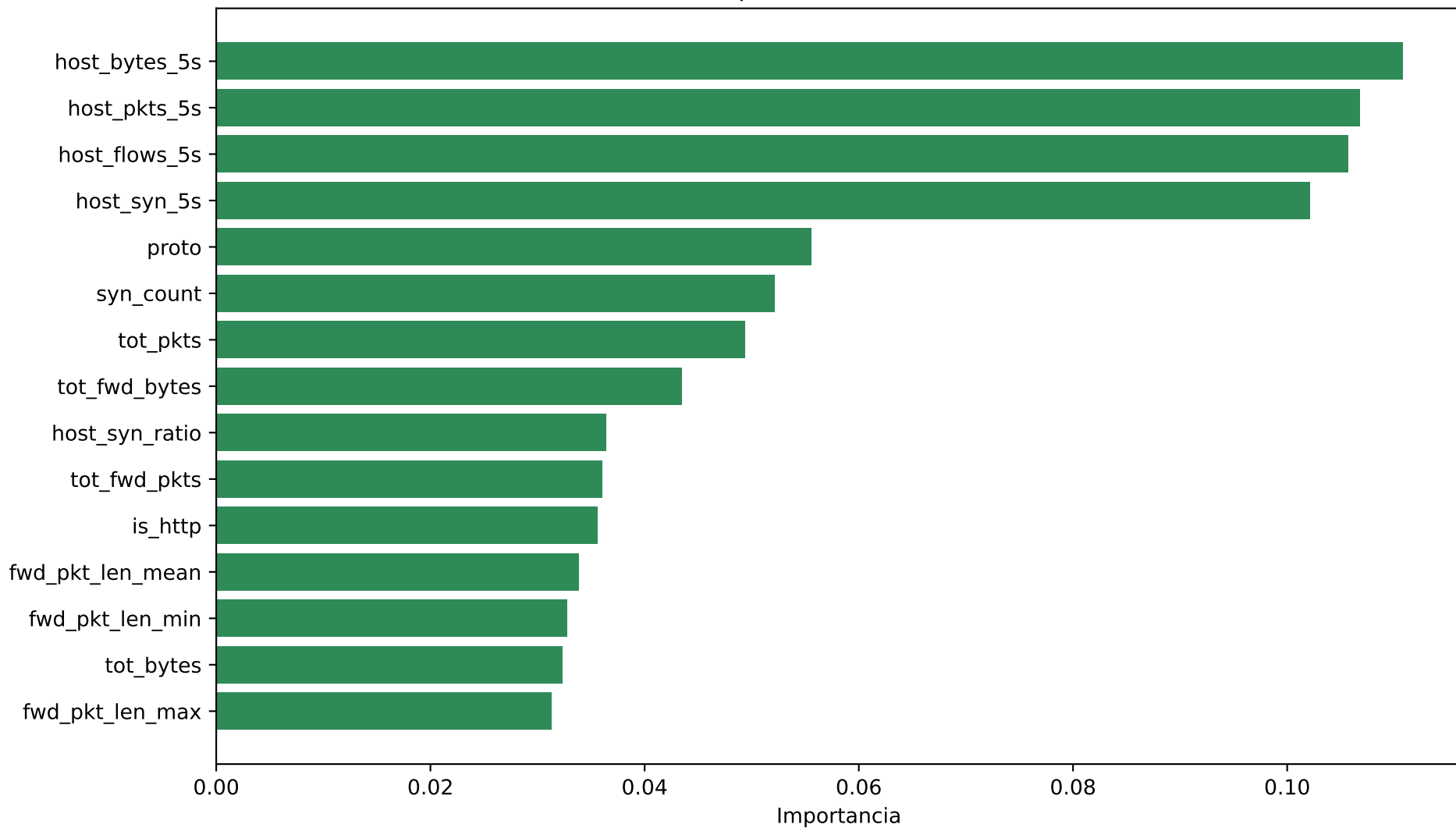
Matriz de confusión — rf



ROC OvR — rf



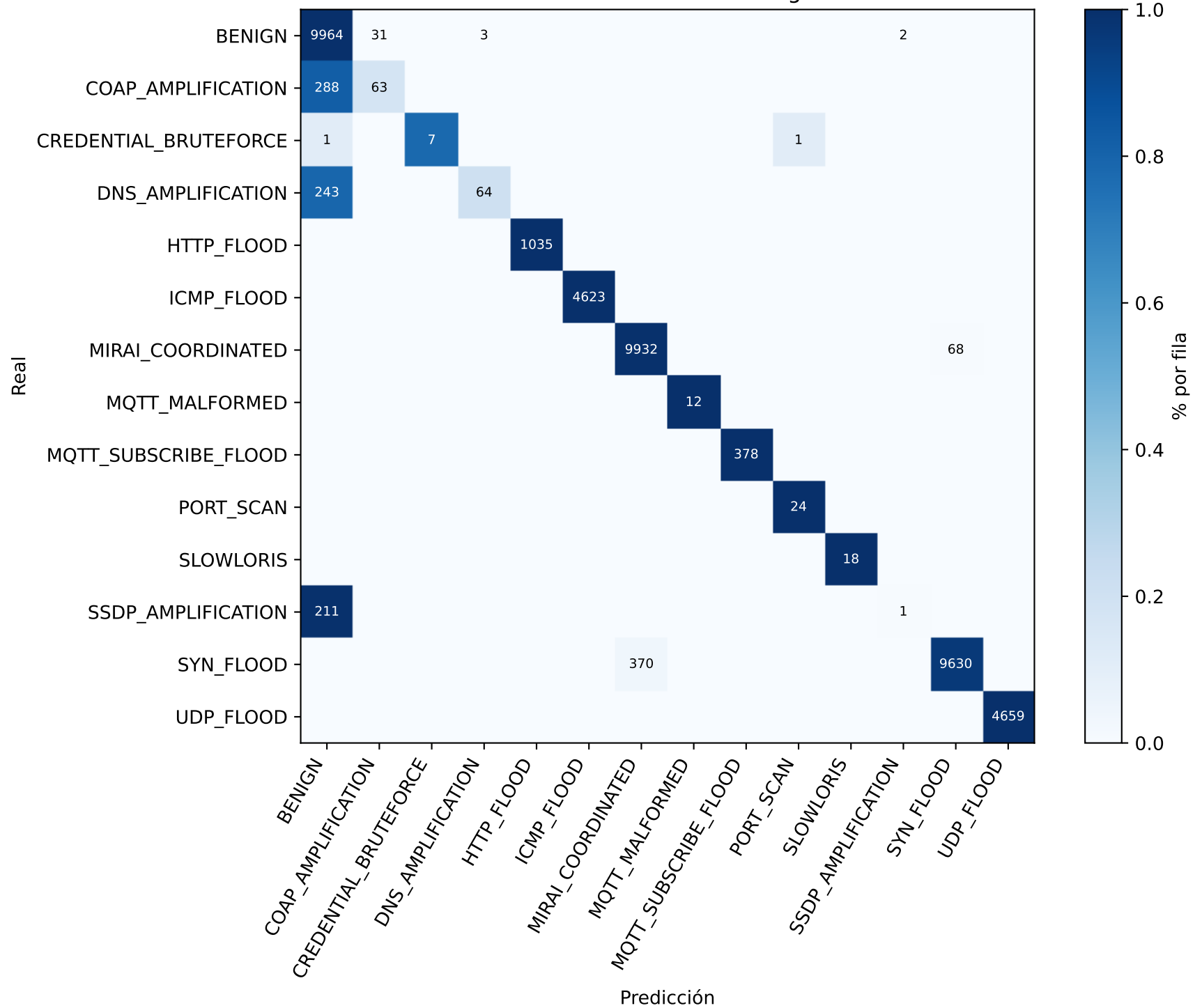
Top-15 features — rf



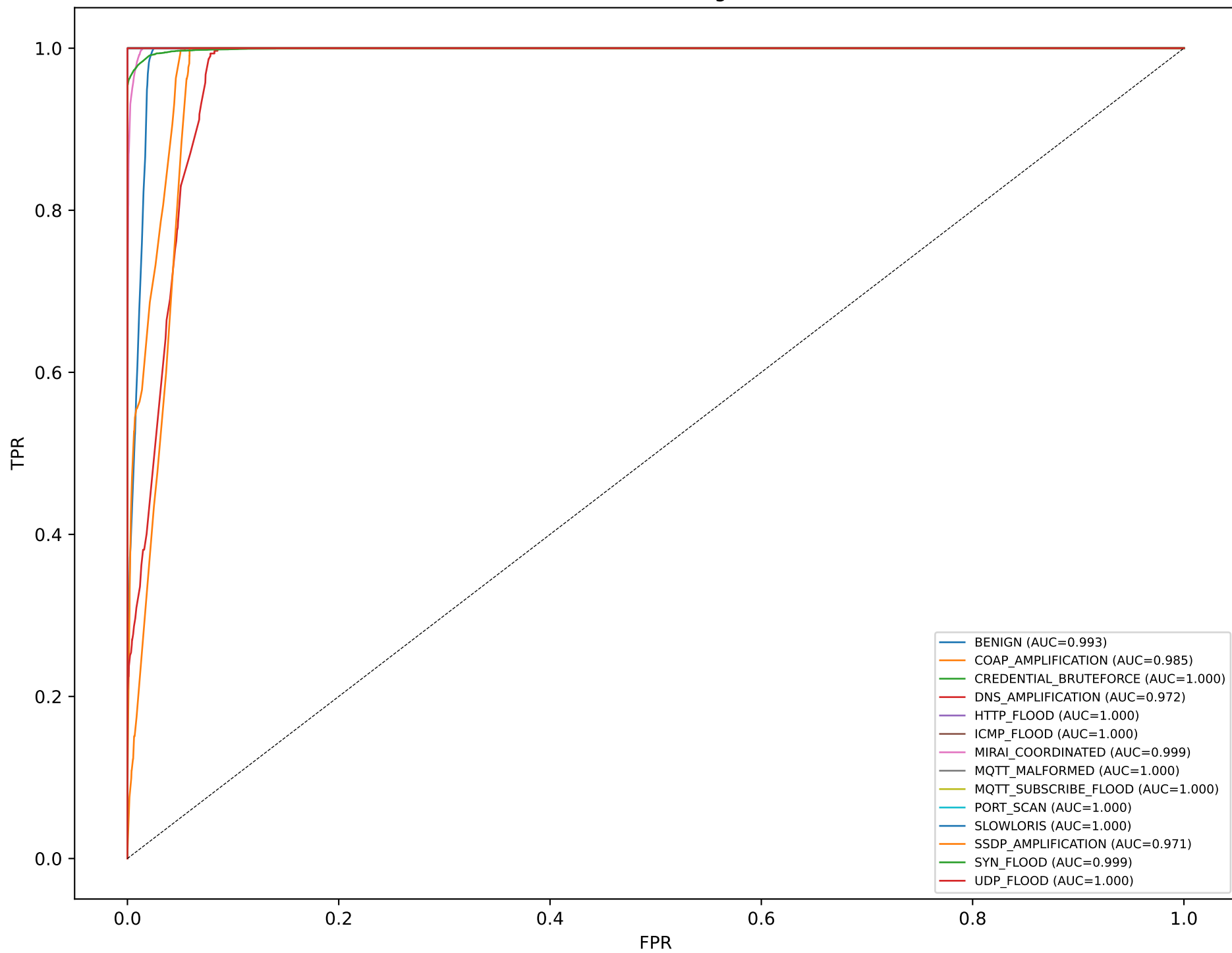
Classification report — xgb

	precision	recall	f1-score	support
BENIGN	0.93	1.00	0.96	10000
COAP_AMPLIFICATION	0.67	0.18	0.28	351
CREDENTIAL_BRUTEFORCE	1.00	0.78	0.88	9
DNS_AMPLIFICATION	0.96	0.21	0.34	307
HTTP_FLOOD	1.00	1.00	1.00	1035
ICMP_FLOOD	1.00	1.00	1.00	4623
MIRAI_COORDINATED	0.96	0.99	0.98	10000
MQTT_MALFORMED	1.00	1.00	1.00	12
MQTT_SUBSCRIBE_FLOOD	1.00	1.00	1.00	378
PORT_SCAN	0.96	1.00	0.98	24
SLOWLORIS	1.00	1.00	1.00	18
SSDP_AMPLIFICATION	0.33	0.00	0.01	212
SYN_FLOOD	0.99	0.96	0.98	10000
UDP_FLOOD	1.00	1.00	1.00	4659
accuracy			0.97	41628
macro avg	0.91	0.79	0.81	41628
weighted avg	0.97	0.97	0.96	41628

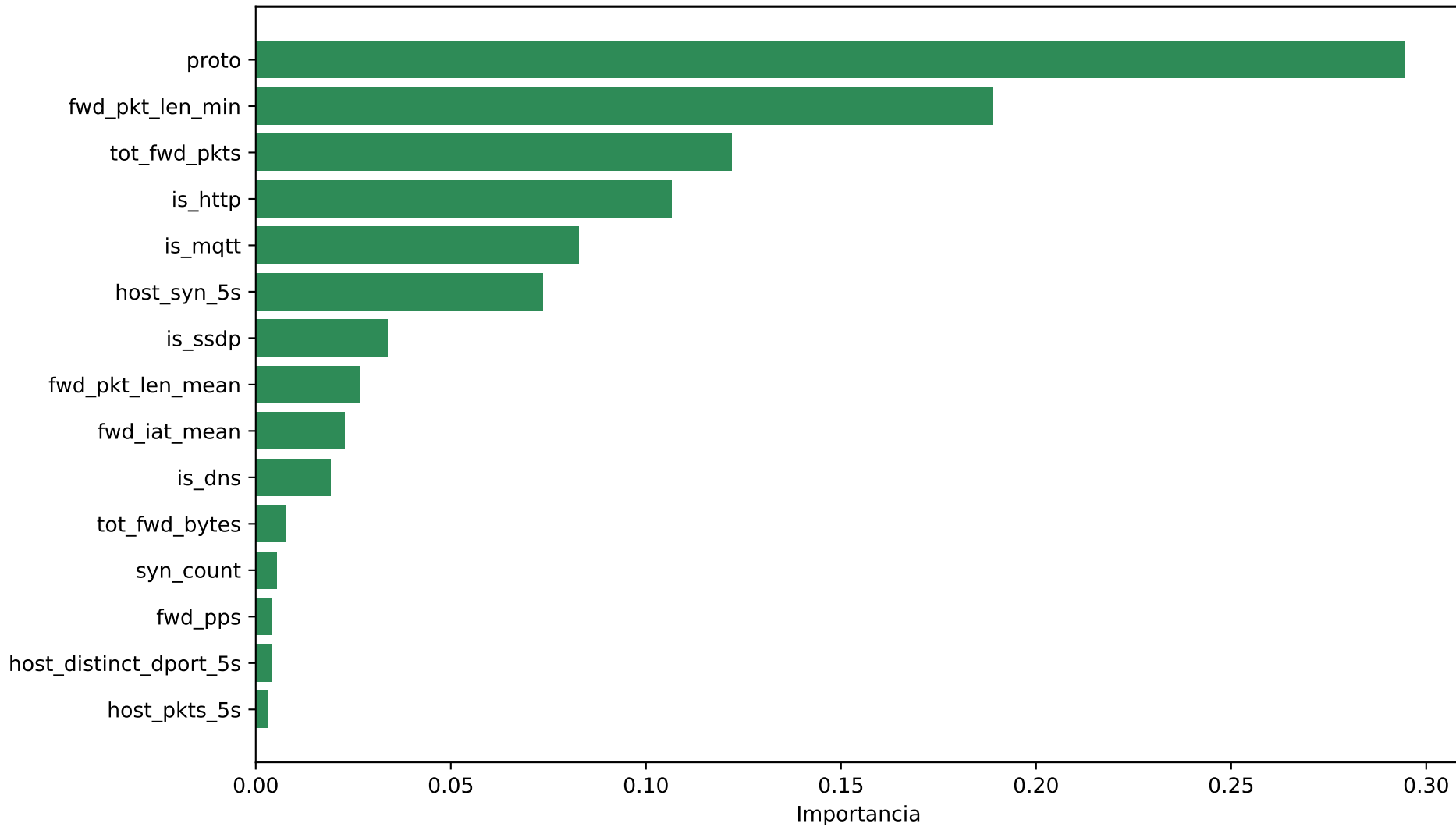
Matriz de confusión — xgb



ROC OvR — xgb



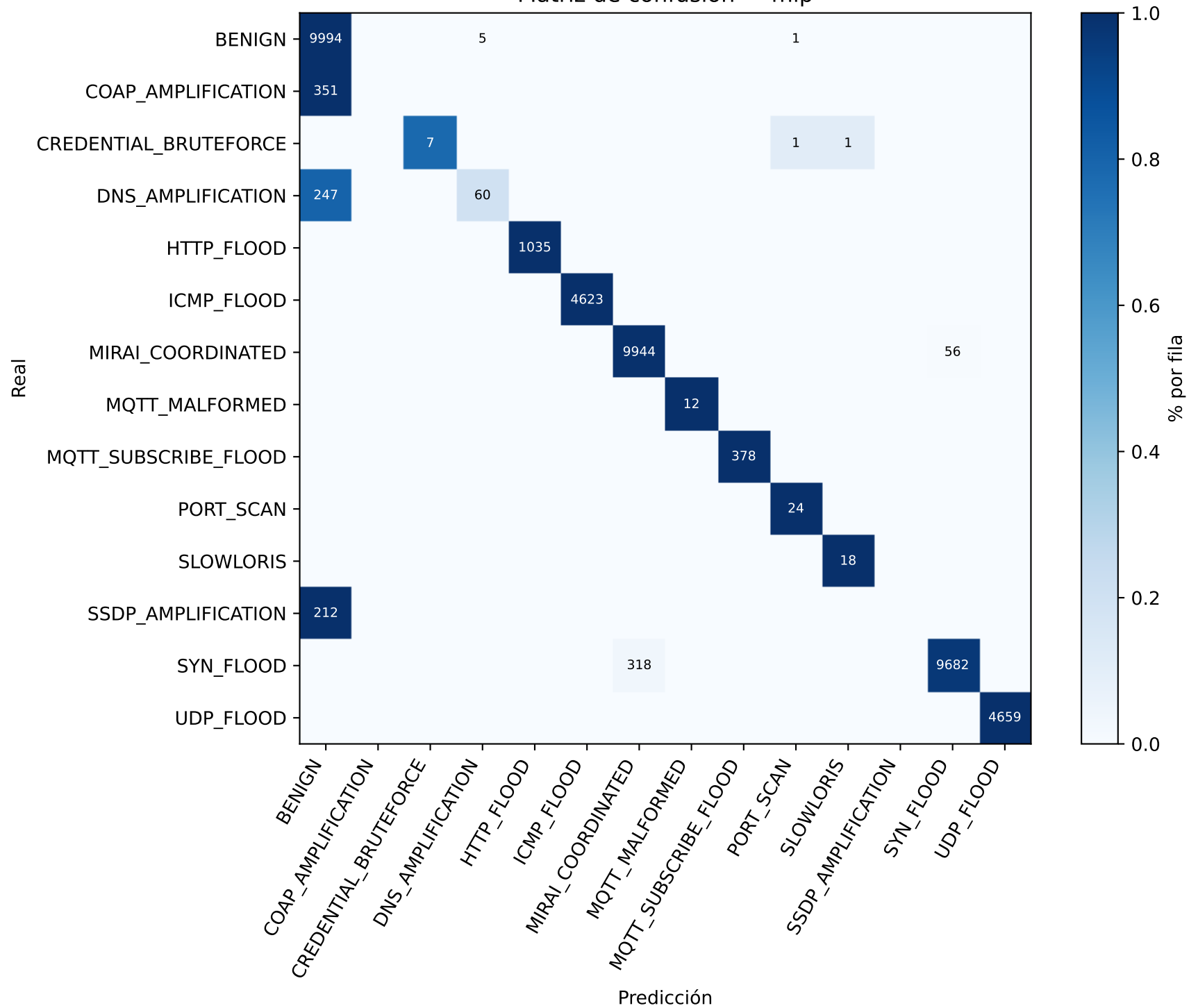
Top-15 features — xgb



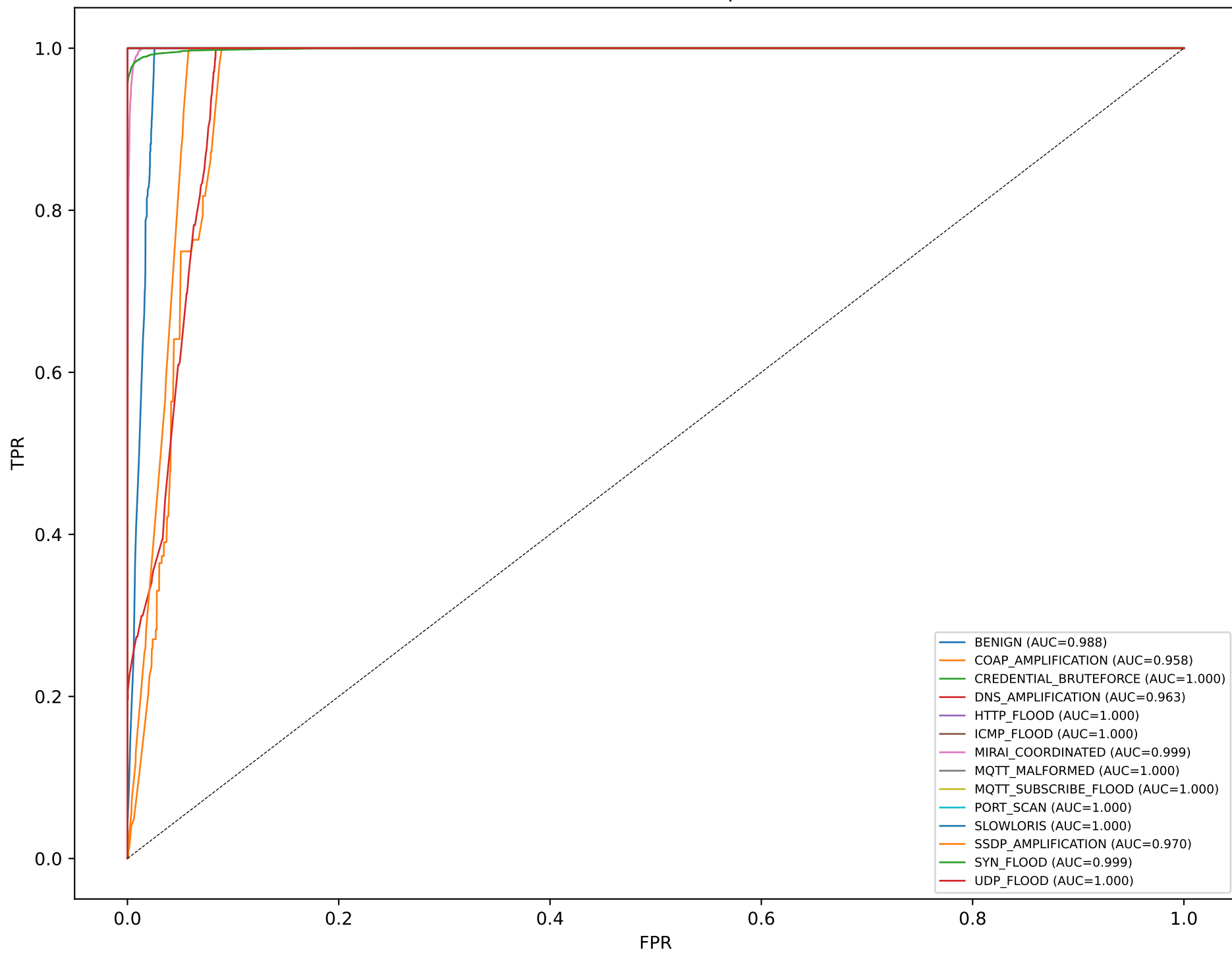
Classification report — mlp

	precision	recall	f1-score	support
BENIGN	0.93	1.00	0.96	10000
COAP_AMPLIFICATION	0.00	0.00	0.00	351
CREDENTIAL_BRUTEFORCE	1.00	0.78	0.88	9
DNS_AMPLIFICATION	0.92	0.20	0.32	307
HTTP_FLOOD	1.00	1.00	1.00	1035
ICMP_FLOOD	1.00	1.00	1.00	4623
MIRAI_COORDINATED	0.97	0.99	0.98	10000
MQTT_MALFORMED	1.00	1.00	1.00	12
MQTT_SUBSCRIBE_FLOOD	1.00	1.00	1.00	378
PORT_SCAN	0.92	1.00	0.96	24
SLOWLORIS	0.95	1.00	0.97	18
SSDP_AMPLIFICATION	0.00	0.00	0.00	212
SYN_FLOOD	0.99	0.97	0.98	10000
UDP_FLOOD	1.00	1.00	1.00	4659
accuracy			0.97	41628
macro avg	0.83	0.78	0.79	41628
weighted avg	0.96	0.97	0.96	41628

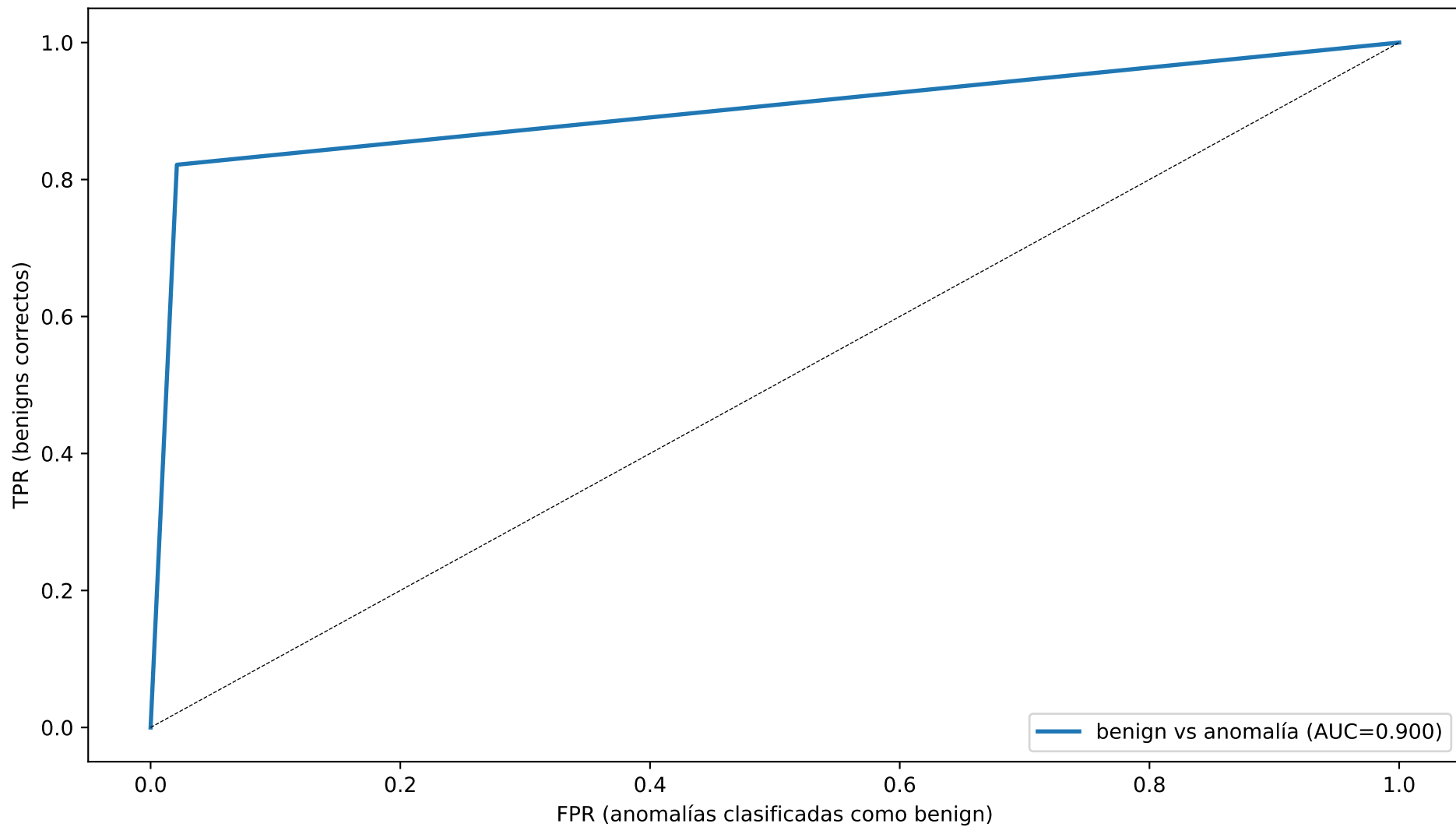
Matriz de confusión — mlp



ROC OvR — mlp



ROC binaria — isoforest



ROC binaria — autoencoder

